

KUEMMERSBRUCK, Germany

WMO: 107710

Lat: 49.4283N

Lon: 11.9017E

Elev: 418

StdP: 96.40

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.1	-15.0	1.1	-12.6	-12.2	1.4	-9.1	10.7	5.4	9.3	5.1	1.1	100	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	30.8	19.0	28.7	18.3	26.8	17.6	20.0	28.0	19.1	26.7	18.3	25.1	2.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.0	21.6	16.5	12.4	21.2	15.7	11.8	20.9	58.8	27.9	55.8	26.8	53.2	25.3	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	6.7	5.6	-16.9	33.5	3.5	1.8	-19.4	34.7	-21.5	35.8	-23.5	36.8	-26.1	38.0
			-16.6	21.4	3.6	1.3	-19.2	22.3	-21.3	23.1	-23.3	23.8	-25.9	24.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-1.2	0.4	4.1	8.8	13.5	16.8	18.3	18.1	13.6	8.7	3.6	-0.2
	DBStd	8.00	4.62	4.52	4.08	4.00	3.72	3.84	3.43	3.22	3.34	3.82	3.51	4.28
	HDD10.0	1468	348	269	184	68	10	1	0	0	7	71	194	315
	HDD18.3	3616	607	503	441	287	157	75	44	44	144	298	443	573
	CDD10.0	1012	0	0	2	32	117	204	258	251	115	31	1	0
	CDD18.3	117	0	0	0	0	6	28	44	36	3	0	0	0
	CDH23.3	1446	0	0	0	9	106	333	492	446	59	1	0	0
	CDH26.7	446	0	0	0	1	21	100	172	146	7	0	0	0
Wind	WSAvg	2.2	2.4	2.6	2.7	2.3	2.2	2.1	2.2	2.0	2.0	1.9	2.1	2.3
Precipitation	PrecAvg	753	61	52	52	43	69	79	81	70	62	56	63	65
	PrecMax	1110	135	136	143	92	159	147	133	146	163	197	146	184
	PrecMin	516	8	8	10	11	18	21	18	23	23	14	1	12
	PrecStd	118	33	32	34	21	34	32	32	31	31	36	31	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	12.5	18.1	24.4	29.2	32.1	33.2	34.5	27.7	21.7	14.2	10.4
		MCWB	8.0	7.8	10.2	13.5	17.4	19.5	19.7	19.4	17.3	15.5	10.8	8.3
	2%	DB	8.9	9.9	15.5	21.9	26.2	29.5	30.9	30.1	25.1	18.4	11.7	8.7
		MCWB	6.8	6.4	9.2	13.1	16.6	18.8	19.0	18.5	17.2	13.7	9.3	6.9
	5%	DB	7.1	8.2	12.9	19.4	23.7	27.4	28.5	27.8	22.8	16.4	10.0	6.7
		MCWB	5.3	5.7	8.2	11.7	15.1	17.9	18.4	17.8	15.7	12.8	8.1	5.3
	10%	DB	5.1	6.7	10.6	17.0	21.3	24.8	25.9	25.7	20.5	14.5	8.4	5.1
		MCWB	3.7	4.5	7.2	10.6	13.9	16.7	17.7	17.3	14.9	11.6	7.0	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	8.4	11.2	14.9	18.6	20.7	20.9	21.0	19.3	15.9	11.2	8.6
		MCDB	10.2	11.4	15.8	22.8	27.3	29.5	29.8	29.9	24.9	20.1	13.3	10.0
	2%	WB	6.9	7.0	9.9	13.4	17.0	19.5	19.9	19.6	17.6	14.4	9.7	7.0
		MCDB	8.6	9.4	14.2	20.8	25.0	27.4	28.2	27.3	23.1	17.7	11.4	8.5
	5%	WB	5.4	5.8	8.7	12.1	15.7	18.5	19.1	18.7	16.5	13.1	8.3	5.4
		MCDB	7.0	8.0	12.3	18.5	22.4	25.7	26.7	25.9	21.5	15.8	9.7	6.6
	10%	WB	3.7	4.5	7.4	10.9	14.6	17.4	18.2	17.8	15.4	11.9	7.0	4.1
		MCDB	5.0	6.5	10.4	16.1	20.3	23.7	24.6	24.2	19.7	14.1	8.3	5.0
Mean Daily Temperature Range	5% DB	MDBR	5.1	6.7	8.6	11.2	11.4	11.3	11.3	11.4	10.8	8.4	5.2	4.6
		MCDBR	6.1	9.2	13.5	16.2	16.1	15.8	16.2	16.4	15.4	11.2	7.9	6.2
		MCWBR	5.0	6.4	8.1	8.4	7.8	6.8	6.3	6.8	7.9	6.9	5.8	5.0
	5% WB	MCDBR	5.9	8.1	11.6	15.0	14.4	14.0	13.8	13.8	13.3	10.0	7.2	5.9
		MCWBR	5.0	5.9	7.3	8.0	7.3	6.4	5.6	6.1	7.1	6.5	5.5	4.9
Clear-Sky Solar Irradiance	taub	0.300	0.321	0.359	0.393	0.399	0.400	0.410	0.404	0.375	0.351	0.325	0.297	
	taud	2.421	2.371	2.329	2.269	2.282	2.306	2.276	2.314	2.370	2.431	2.460	2.433	
	Ebn at Noon	756	822	846	851	860	860	844	832	819	777	718	706	
	Edh at Noon	63	84	104	123	127	125	127	116	99	79	62	55	
All-Sky Solar Radiation	RadAvg	0.87	1.72	2.79	4.24	4.99	5.51	5.23	4.58	3.30	1.98	0.96	0.68	
	RadStd	0.15	0.27	0.36	0.55	0.57	0.46	0.51	0.45	0.44	0.24	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
						+0.30	+0.39	N/A	N/A	+57	+21

Nomenclature: See separate page